DTU Compute
Department of Applied Mathematics and Computer Science

Occupancy-Aware Climate Control Estimating Thermal Comfort Curves with Hidden Markov Models

Bachelor Thesis



Occupancy-Aware Climate Control
Estimating Thermal Comfort Curves with Hidden Markov Models

Bachelor Thesis
7 December, 2026

By
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Approval

This thesis has been prepared over six months at the Section for Indoor Climate, Department of Civil Engineering, at the Technical University of Denmark, DTU, in partial fulfilment for the degree Master of Science in Engineering, MSc Eng.

It is assumed that the reader has a basic knowledge in the areas of statistics.

Mads Holme Håkonsson - s234936

.....

Signature

.....

Date

Abstract

ABSTRACT IS MISSING

Acknowledgements

Mads Holme Håkonsson, MSc Civil Engineering, DTU
Creator of this thesis template.

[Name], [Title], [affiliation]
[text]

[Name], [Title], [affiliation]
[text]

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1 Introduction

2 Colours

The design guide define 3 primary colours (dtured, white and black) and 10 secondary colours https://www.designguide.dtu.dk/#stnd-colours. Below are codes for the various colour modes. RGB is used for web and Office Programmes. CMYK is used for print. HTML is used for HTML-coding. If you know anything about colour codes you might notice that the RGB codes are ranging from 0-1 instead of the usual 0-255.

color	rgb	cmyk	HTML
dtured	0.77 0 0.05	0 0.91 0.72 0.23	C4000D
white	1 1 1	0 0 0 0	FFFFFF
black	0 0 0	0 0 0 1	000000
blue	0.12 0.24 1	0.88 0.76 0 0	1F3DFF
brightgreen	0.31 1 0.34	0.69 0 0.66 0	4FFF57
navyblue	0 0 0.4	1 0.9 0 0.6	000066
yellow	0.95 0.83 0.18	0.05 0.17 0.82 0	F2D42E
orange	1 0.35 0.14	0 0.65 0.86 0	FF5924
pink	1 0.65 0.74	0 0.35 0.26 0	FFA6BD
red	1 0.14 0.35	0 0.86 0.65 0	FF2459
green	0 0.78 0	0.89 0.05 1 0.17	00C700
purple	0.33 0.04 1	0.67 0.96 0 0	540AFF

The default colour mode for this template is cmyk. The current colour model is cmyk which is also illustrated by the underlined numbers in the colour test table above. If you which to change the colour model to rgb go to Setup/Settings.tex and change `targetcolourmodel` to rgb. In Setup/Settings.tex it is also possible to change the background colour of the front and back page. The colours are primarily used for diagrams (the plotcyclelist DTU) and the front and back page.

Lighter colours can be achieved as written in the LaTeX code below. For example to get a tint of 50% you would write colourname!50.
Normal dtured 80% dtured 70% dtured 60% dtured 50% dtured
For more information about colours in LaTeX read the `xcolor` manual.

3 Examples of figures, tables, equations and listings

In the following a bunch of examples of figures and tables have been made. There are advantages to using `tikZ` diagrams over excel diagrams. 1) the font and font size perfectly matches the document 2) the styling and colours are pre-defined to follow the design guide 3) the plots uses vector graphics which reduces the file size, reduces the compile time and looks sharp when zooming in. The possibilities are endless, look at the `pgfplots` gallery for inspiration: `http://pgfplots.sourceforge.net/gallery.html`. However there are still cases where I would recommend to insert a plot as a picture. For example if the plot contains a lot of data: a line graph with 1000 points takes a long time to compile.

Some tips if you want good looking diagrams or graphs which will be inserted as pictures (e.g. in a figure environment with \includegraphics): The main font is Arial. Use DTU colours as described in chapter 2. Use high quality pictures. Try to scale the diagram (picture) so the text size of the axis legends match the text size in this document.

Remember to change the label of your figures so there are no duplicate labels. A label should be placed below a caption or after a heading (fx after a \chapter).

3.1 Graphs and charts

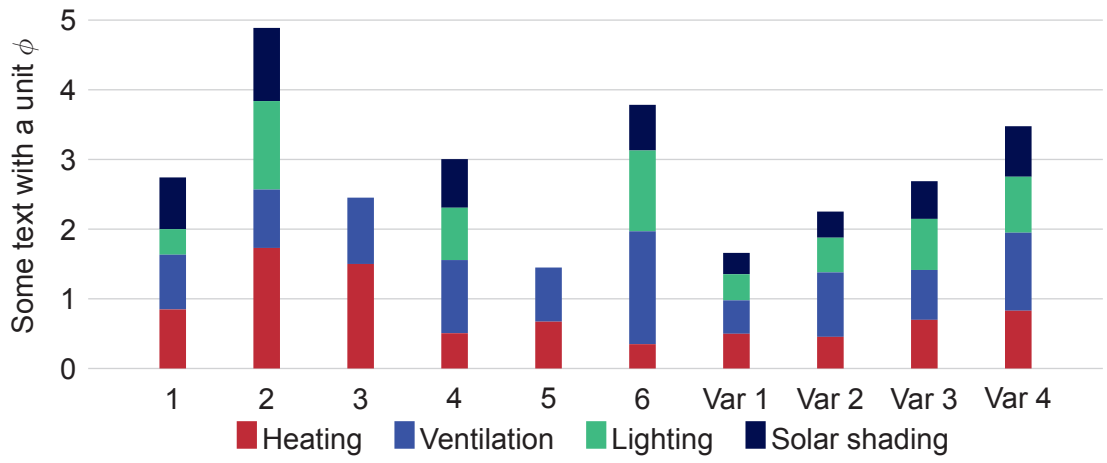


Figure 3.1: Stacked column chart

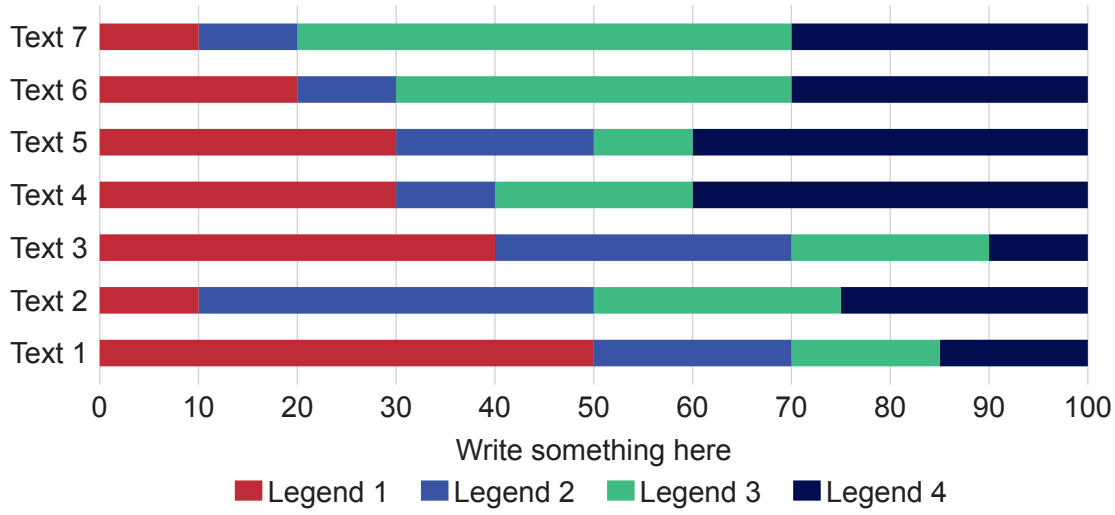


Figure 3.2: Stacked bar chart

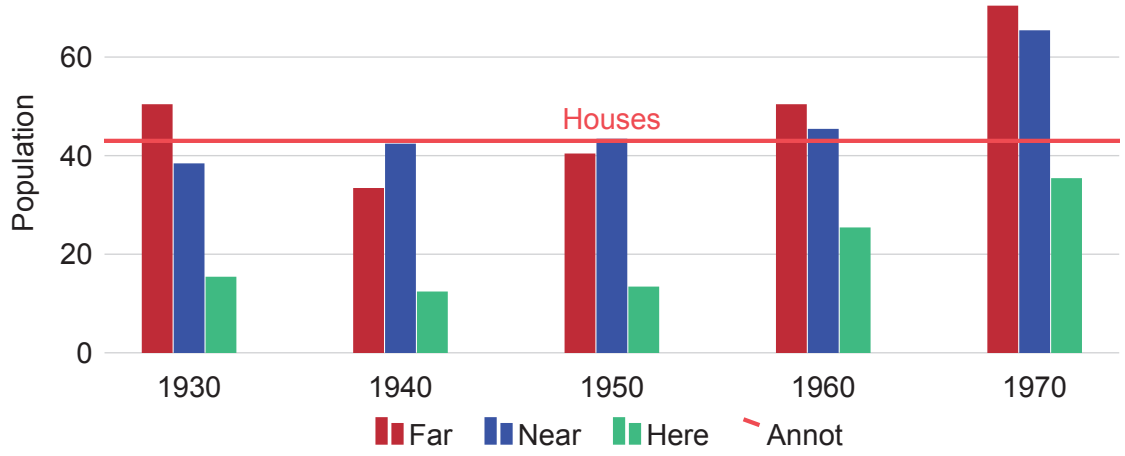


Figure 3.3: Grouped column chart

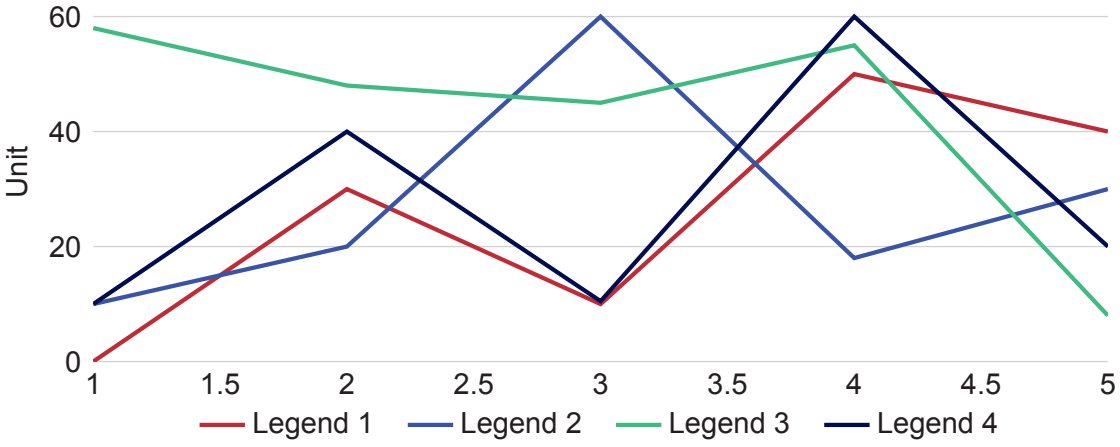


Figure 3.4: Line graph

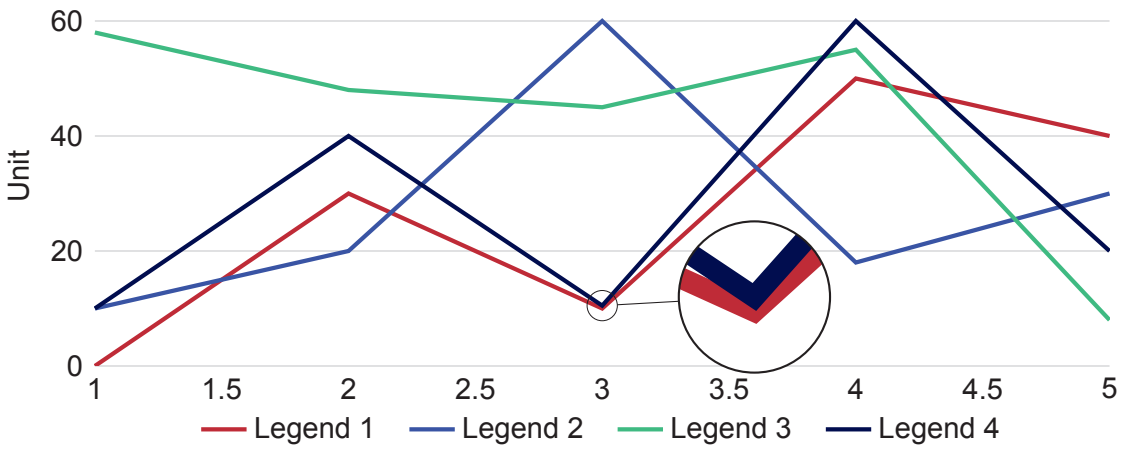


Figure 3.5: Line graph with magnifying glass

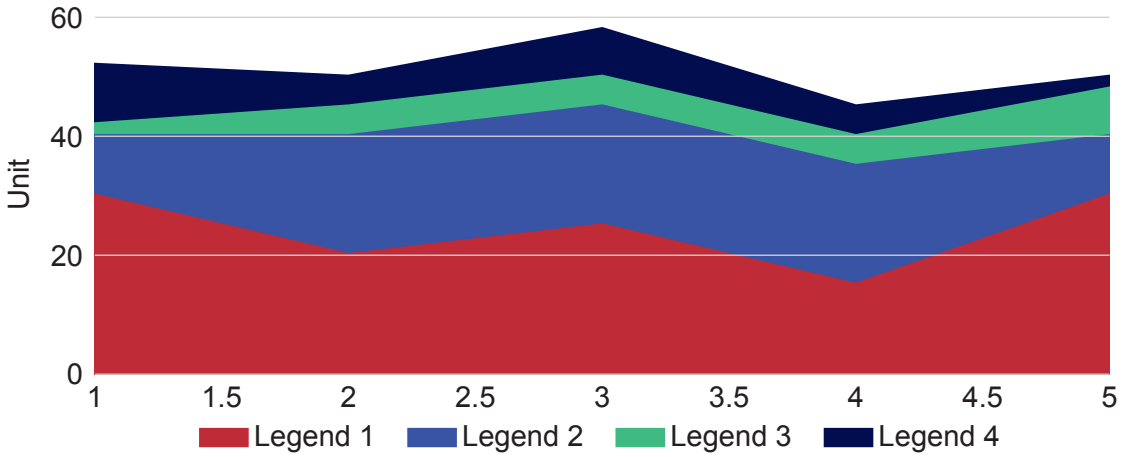


Figure 3.6: Area graph

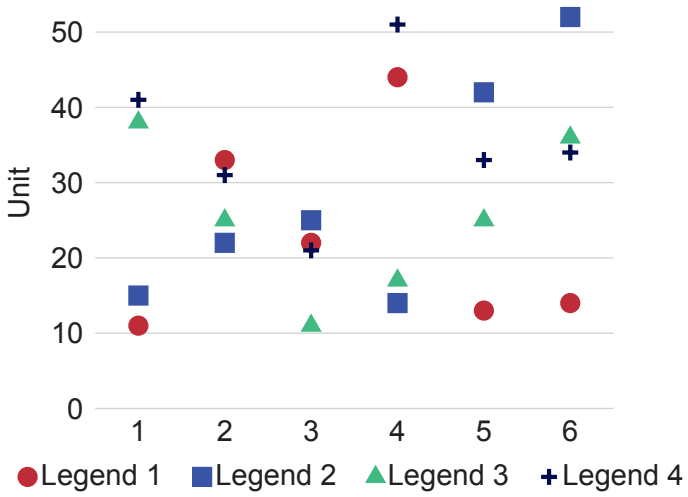


Figure 3.7: Scatter plot

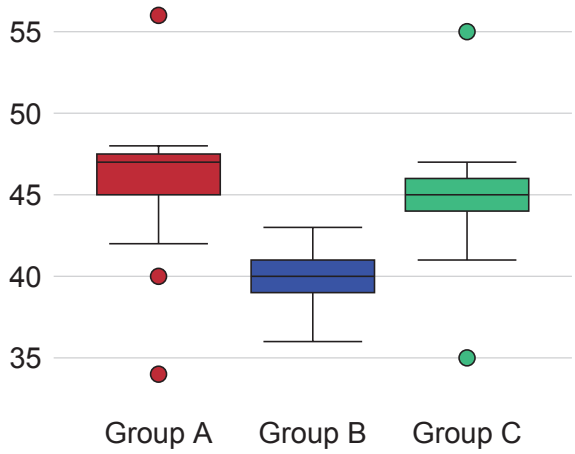


Figure 3.8: Boxplot

3.2 Tables and figures

Table 3.1: This is a `booktabs` table. Go to `http://www.tablesgenerator.com/` and use the booktabs table style

	Item	
Animal	Description	Price ($)
Gnat	per gram	13.65
	each	0.01
Gnu	stuffed	92.50
Emu	stuffed	33.33
Armadillo	frozen	8.99

`Booktabs` tables don't use any vertical lines. Only horizontal lines are used. Table 3.1 begins with a `\toprule`, ends with a `\bottomrule` with `\midrule` in between. The table has 3 columns formatted as `@{}llS@{}`. `@{}` is cropping the horizontal lines of the table to fit the content (removes column spacing at the left and right edges). `l` aligns the column to the left and `S` aligns the column according to the decimal point (`siunitx` package). You can of course also use `r` to align right or `c` to center the contents of the column.

Table 3.2: Wrongly formatted table

	Voltage V	Current A	Power W
Transformer input	234.4	0.50	117.4
Transformer output	25.86	2.72	70.3
Efficiency			60%

Table 3.3: Correctly formatted table

	Voltage V	Current A	Power W
Transformer input	234.4	0.50	117.4
Transformer output	25.86	2.72	70.3
Efficiency			60 %

Table 3.2 and table 3.3 have the same comtents but there are some subtle differences in formatting which makes table 3.3 the superior table of the two. The most obvious change is removing the midrule between the transformer input and output rows. The efficency row is the odd man out and a midrule has been used to emphasise the difference between the transformer rows and the efficiency row. The delimiters in the voltage, current and power columns are aligned. The horizontal lines (rules) fits to the content and instead of protruding. The spacing between 60 and the percentage sign is correctly adjusted.



Figure 3.9: Just a normal figure



(a) A subfigure (b) A subfigure

Figure 3.10: A figure with two subfigures



(a) A subfigure



(b) A subfigure



(c) A subfigure



(d) A subfigure

Figure 3.11: A figure with four subfigures

Referring to the figure as a whole fig. 3.11 or to an individual sub figure fig. 3.11a is done the normal way with `\cref{}` commands.

3.3 Equations

In-line math is easy. Anything surrounded by dollar signs becomes a math field. Here is an example: $f(x) = 2x - 1$. Also anything inside the "\begin{equation}" and "\end{equation}" environment is also a math field. Examples are shown below.

All equations use the default latex font. Some might say it looks weird with a serif font for equations and a sans-serif font for the body text. However, it is very unpractical to change the math font in latex which is the exactly the reason why this has not been done. One benefit of the serif style math font is the clear distinction between symbols (variables) and units.

On the subject of units, those are all taken care of by the `\siunitx` package. Whenever there is a number followed by a unit one should write \SI{number}{unit}. Note this command is case sensitive. If a unit should follow a variable use the command \si{unit} (also case sensitive).

The ideal gas law is shown in eq. (3.1).

$$p \cdot V = n \cdot R \cdot T \tag{3.1}$$

$$\frac{\partial}{\partial t} \int_0^\delta U dy = -\delta \frac{1}{\rho} \frac{\partial P}{\partial x} - U_f(t)^2 \tag{3.2}$$

$$d_{step} = \sqrt{\frac{\delta}{\frac{dw}{dp_v}} \cdot t} = \sqrt{\frac{1.0 \times 10^{-11}\,\mathrm{kg/(m\,s\,Pa)}}{\frac{5.4\,\mathrm{kg/m^3}}{233.82\,\mathrm{Pa}}} \cdot 7200\,\mathrm{s}} = 0.001\,766\,\mathrm{m} = 1.766\,\mathrm{mm} \tag{3.3}$$

$$x = \mathtt{x}, \mathbf{x}, \boldsymbol{x}, x_{1_{2_{3_4}}}^{1^{2^{3^4}}} \cdot hello * \text{hello world} \cdot \text{equation without number}$$

Notice how the `aligned` environment can be used to align the equilibrium arrows in eq. (3.4). Only one equation number is generated using this method. Alternatively if you want an equation number for each line see eqs. (3.5) to (3.6).

$$\begin{aligned} CH_3COOH + OH^- &\rightleftharpoons CH_3COO^- + H_2O \\ H_2O &\rightleftharpoons H^+_{(aq)} + OH^-_{(aq)} \end{aligned} \tag{3.4}$$

$$f(x) = 1 + x - 3x^2 \tag{3.5}$$

$$g(x) + y = 3x - \frac{1}{2}x^3 \tag{3.6}$$

3.4 Listings (code)

Listing 3.1 is a nicely formatted block of code. A listing will automatically continue on the next page if it encounters a page break. Many different programming languages can be highlighted. Check the `listings` package documentation for a list of supported programming languages.

```
%% Monte Carlo simulation, estimation of pi
m=1E7;

x=rand(m,1);
y=rand(m,1);

g = x.^2+y.^2-1;

%dots outside
Pf = sum((g)<=0)/m

pi = 4*Pf
```

Listing 3.1: Monte Carlo simulation to estimate the value of π

A Title

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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